Experiment 9: Property Data Analysis Using Pandas

Aim

To find the average listing price by location, count properties with more than four bedrooms, and identify the property with the largest area using Pandas.

Algorithm

1. Import Pandas.
2. Get property details as input.
3. Calculate the average price for each location.
4. Count properties with more than four bedrooms.
5. Find the property with the largest area.

Code

Experiment 9 Code

```
import pandas as pd

id = input("Enter property IDs: ").split()
loc = input("Enter locations: ").split()
bed = list(map(int, input("Enter bedrooms: ").split()))
area = list(map(float, input("Enter areas: ").split()))
price = list(map(float, input("Enter prices: ").split()))

df = pd.DataFrame({"ID": id, "Location": loc, "Bedrooms": bed,
"Area": area, "Price": price})

print(df.groupby("Location")["Price"].mean())
print("More than 4 bedrooms:", (df["Bedrooms"] > 4).sum())
print("Largest property:\n", df.loc[df["Area"].idxmax()])
```

Input

Enter property IDs: P1 P2 P3

Enter locations: Chennai Madurai Chennai

Enter bedrooms: 3 5 6

Enter areas: 1200 2000 2500

Enter prices: 3000000 4000000 5000000

OUTPUT

```
================ RESTART: C:/Users/arunp/OneDrive/Documents/FDS/EXP 9.py ==============
Enter property IDs: P1 P2 P3
Enter locations: Chennai Hosur Krishnagiri
Enter bedrooms: 3 5 6
Enter areas: 1200 2000 2500
Enter prices: 3000000 4000000 5000000
Location
Chennai        3000000.0
Hosur          4000000.0
Krishnagiri    5000000.0
Name: Price, dtype: float64
More than 4 bedrooms: 2
Largest property:
 ID                   P3
Location    Krishnagiri
Bedrooms              6
Area             2500.0
Price         5000000.0
Name: 2, dtype: object
```

Result

Thus, the average listing price by location, the number of properties with more than four bedrooms, and the property with the largest area were successfully identified using Pandas.